

4-4a Equilibrium

Figure 8 shows the market supply curve and market demand curve together. Notice that there is one point at which the supply and demand curves intersect. This point is called the market's

equilibrium (a situation in which the market price has reached the level at which quantity supplied equals quantity demanded)

. The price at this intersection is called the

equilibrium price (the price that balances quantity supplied and quantity demanded) , and the quantity is called the

equilibrium quantity (the quantity supplied and the quantity demanded at the equilibrium price)

. Here the equilibrium price is \$2.00 per cone, and the equilibrium quantity is 7 ice-cream cones.

Figure 8

The Equilibrium of Supply and Demand

The equilibrium is found where the supply and demand curves intersect. At the equilibrium price, the quantity supplied equals the quantity demanded. Here the equilibrium price is \$4: At this price, 7 ice-cream cones are supplied and 7 ice-cream cones are demanded.

The dictionary defines the word *equilibrium* as a situation in which various forces are in balance. This definition applies to a market's equilibrium as well. *At the equilibrium price, the*

quantity of the good that buyers are willing and able to buy exactly balances the quantity that sellers are willing and able to sell. The equilibrium price is sometimes called the *market-clearing price* because, at this price, everyone in the market has been satisfied: Buyers have bought all they want to buy, and sellers have sold all they want to sell.

The actions of buyers and sellers naturally move markets toward the equilibrium of supply and demand. To see why, consider what happens when the market price is not equal to the equilibrium price.

Suppose first that the market price is above the equilibrium price, as in panel (a) of [Figure 9](#). At a price of \$5 per cone, the quantity of the good supplied (10 cones) exceeds the quantity demanded (4 cones). There is a surplus (a situation in which quantity supplied is greater than quantity demanded) of the good: Producers are unable to sell all they want at the going price. A surplus is sometimes called a situation of *excess supply*. When there is a surplus in the ice-cream market, sellers of ice cream find their freezers increasingly full of ice cream they would like to sell but cannot. They respond to the surplus by cutting their prices. Falling prices, in turn, increase the quantity demanded and decrease the quantity supplied. These changes represent movements *along* the supply and demand curves, not shifts in the curves. Prices continue to fall until the market reaches the equilibrium.

Figure 9

Markets Not in Equilibrium

In panel (a), there is a surplus. Because the market price of \$5 is above the equilibrium price, the quantity supplied (10 cones) exceeds the quantity demanded (4 cones). Producers try to increase sales by cutting the price of a cone, which moves the price toward its equilibrium level. In panel (b), there is a shortage. Because the market price of \$3 is below the equilibrium price, the quantity demanded (10 cones) exceeds the quantity supplied (4 cones). With too many buyers chasing too few goods, producers can take advantage of the shortage by raising the price. Hence, in both cases, the price adjustment moves the market toward the equilibrium of supply and demand.

Suppose now that the market price is below the equilibrium price, as in panel (b) of [Figure 9](#). In this case, the price is \$3 per cone, and the quantity of the good demanded exceeds the quantity supplied. There is a

[**shortage** \(a situation in which quantity demanded is greater than quantity supplied\)](#) of the good: Consumers are unable to buy all they want at the going price. A shortage is sometimes called a situation of *excess demand*. When a shortage occurs in the ice-cream market, buyers have to wait in long lines for a chance to buy one of the few cones available. With too many buyers chasing too few goods, sellers can respond to the shortage by raising their prices without losing sales. These price increases cause the quantity demanded to fall and the quantity supplied to rise. Once again, these changes represent movements *along* the supply and demand curves, and they move the market toward the equilibrium.

Thus, regardless of whether the price starts off too high or too low, the activities of the many buyers and sellers automatically push the market price toward the equilibrium price. Once the market reaches its equilibrium, all buyers and sellers are satisfied, and there is no upward or downward pressure on the price. How quickly equilibrium is reached varies from market to market depending on how quickly prices adjust. In most free markets, surpluses and shortages are only temporary because prices eventually move toward their equilibrium levels. Indeed, this phenomenon is so pervasive that it is called the

[**law of supply and demand** \(the claim that the price of any good adjusts to bring the quantity supplied and the quantity demanded for that good into balance\)](#)

: The price of any good adjusts to bring the quantity supplied and quantity demanded of that good into balance.